Introduction to Numerical Analysis

Course: Introduction to Numerical Analysis (NUM201)
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Summary

Introductory note about approximation, floating-point numbers, numerical error, and the bisection method.

Purpose of numerical analysis

- Numerical analysis studies methods for solving mathematical problems approximately on a computer.
- The goal is to produce answers that are accurate enough, stable, and efficient.

Types of error

- Round-off error appears because computers store real numbers with limited precision.
- Truncation error appears when an infinite process is replaced by a finite number of steps.
- Modeling error appears when the mathematical model simplifies the real system.

Bisection method

- The bisection method finds a root by repeatedly cutting an interval in half.
- It requires a continuous function and opposite signs at the interval endpoints.

Quick reference

Error type	Cause	Example
Round-off	Limited precision	0.1 cannot be stored exactly
Truncation	Finite approximation	Stopping an infinite series
Modeling	Simplified system	Ignoring air resistance